DO NOW, AO1 REVISION

# Thermal conductivity and insulation

**Energy**
Paper 1, Combined and Separate
7 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

| | QUESTION | ANSWER | ✓ |
|---|---|---|---|
| 1 ★ | **Describe how thermal conductivity affects the rate of energy transfer through a material.** | The higher the thermal conductivity, the higher the rate of energy transfer by conduction. | ☐ |
| 2 | **Name two things that affect how fast a building cools through its walls.** | The thickness and the thermal conductivity of the walls | ☐ |
| 3 | **Describe how the thickness of a building's walls affects its rate of energy loss.** | The thicker the walls, the lower the rate of energy loss. | ☐ |
| 4 | **Describe how the thermal conductivity of a building's walls affects its rate of energy loss.** | The higher the thermal conductivity, the higher the rate of energy loss. | ☐ |
| 5 | **Explain the purpose of using a good thermal insulator.** | To reduce unwanted energy transfers by heating. | ☐ |
| 6 ★ | **Explain why loft insulation reduces the heating bills of a house.** | Its low thermal conductivity reduces the rate of energy transfer out of the house. | ☐ |
| 7 ★ | **Give three things that affect the rate of energy transfer through a wall.** | Its thickness, its thermal conductivity and the temperature difference across it. | ☐ |

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